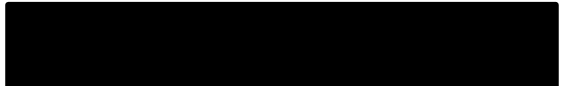
Base conversion (Decimal, binary, octal, hexadecimal)

coaching center



All relevant bases of number systems:

Binary: 0, 1

Octal: 0, 1, 2, 3, 4, 5, 6, 7

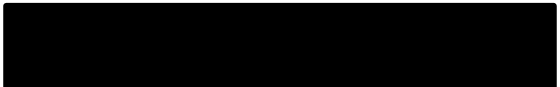
Decimal: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9

Hexadecimal: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F

16

10 11 12 13 14 15

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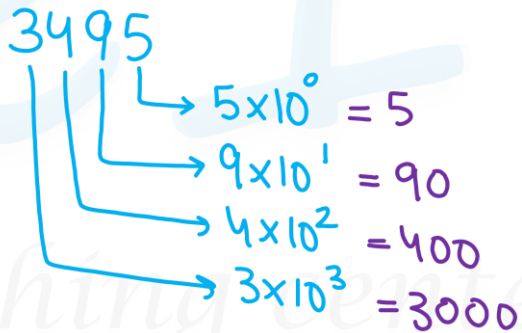


What is base of a number system?

- The number of digits used to create counting

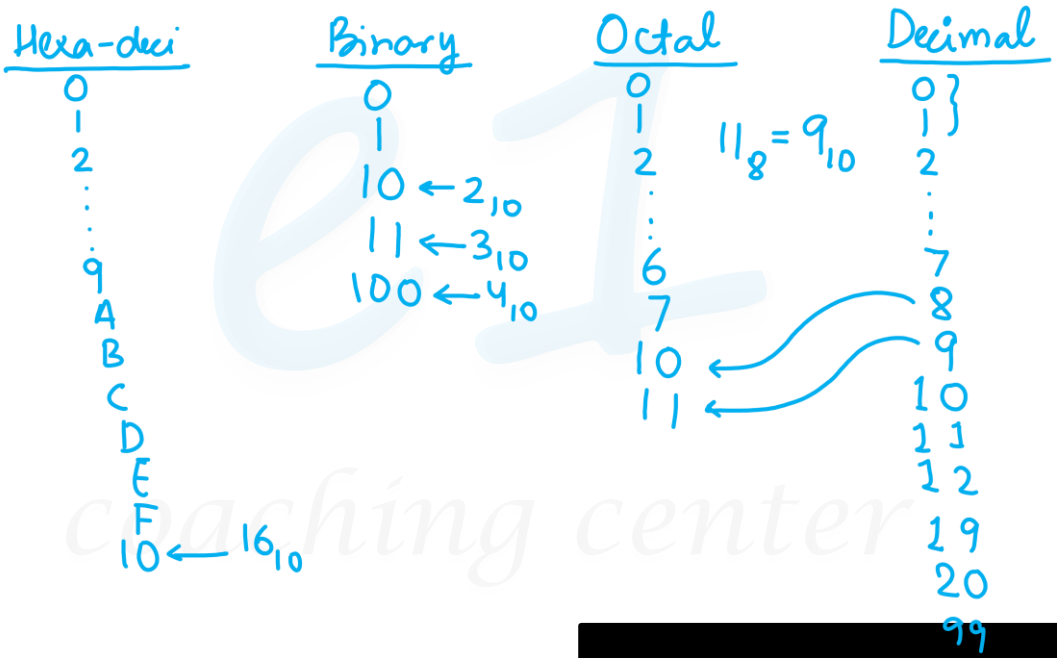
Place value is decided

Example:



coaching center

How are numbers generated?



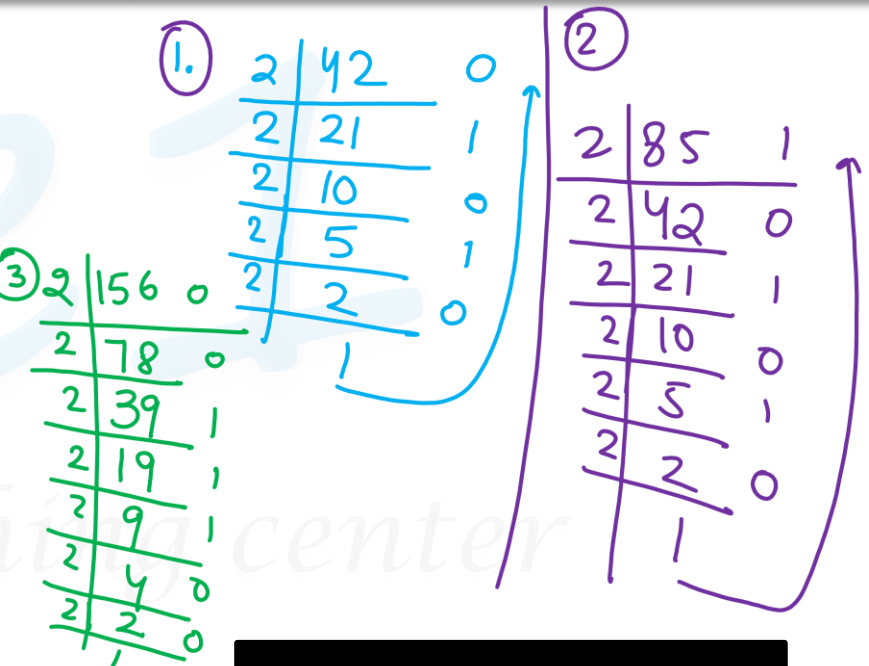
e1 coaching center

Decimal to binary and vice versa

1) $42_{10} = (101010)_2$

2) $85_{10} = (1010101)_2$

3) $156_{10} = (10011100)_2$



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Binary to decimal, last question!

1. $(101010)_2 = (42)_{10}$

$2^5 + 2^3 + 2^1$

$32 + 8 + 2 = 42$

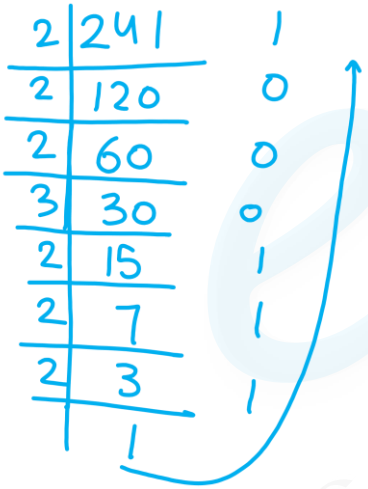
2. $(1010101)_2 = (85)_{10}$

$2^6 + 2^4 + 2^2 + 2^0 = 64 + 16 + 4 + 1 = 85$

3. $(10011100)_2 = (156)_{10}$

$2^7 + 2^4 + 2^3 + 2^2 = 156$

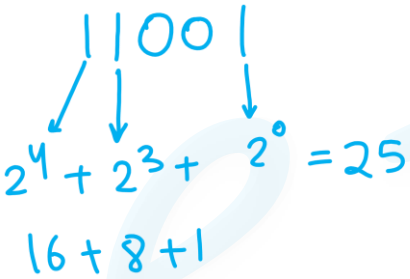
coaching center



1. **Converting the decimal number 241 to the binary number system we get:**
   दशमलव संख्या 241 को बाइनरी संख्या प्रणाली में परिवर्तित करने पर हमें प्राप्त होता है:

   a) 11110000
   b) 11110001
   c) 10110001
   d) 11010001

coaching center



2. The binary number 11001 in the decimal number system is equal to:
दशमलव संख्या प्रणाली में बाइनरी संख्या 11001 इसके बराबर है:

a) 23
b) 24
c) 25
d) 26

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$2^7 + \ldots + 2^1 + 2^0 = 2^8 - 1$

$= 255$

$2^7 + 2^3 = 128 + 8 = 136$

3. **Binary number 11111111 added to 10001000 becomes decimal number:**

बाइनरी संख्या 11111111 को 10001000 में जोड़ने पर दशमलव संख्या बन जाती है:

a) 373

b) 374

c) 391

d) 376

coaching center

$2^7 + \ldots + 2^1 + 2^0 = 2^8 - 1$

$= 255$

$2^3 + 2^2 + 2^1 + 2^0 = 2^4 - 1$

$= 15$

$$\begin{array}{r} 17 \\ 15\overline{)255} \\ \end{array} =$$

4. The binary number 11111111 divided by 1111 gives _____ in the decimal system.
   a) an even number as the answer
   b) a composite number as the answer
   c) the highest prime number less than 20
   d) the second highest prime number less than 20

बाइनरी संख्या 11111111 को 1111 से भाग देने पर दशमलव प्रणाली में _____ प्राप्त होता है।
a) उत्तर के रूप में एक सम संख्या
b) उत्तर के रूप में एक भाज्य संख्या
c) 20 से कम सबसे बड़ी अभाज्य संख्या
d) 20 से कम दूसरी सबसे बड़ी अभाज्य संख्या

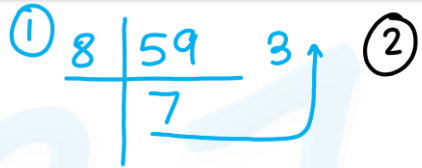
coaching center

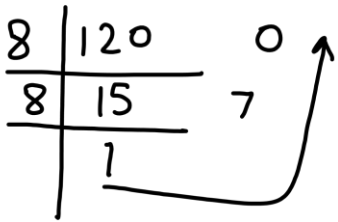
Decimal to Octal and vice versa

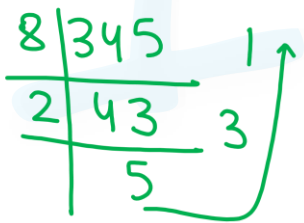
1) $59_{10} = 73_8$

2) $120_{10} = 170_8$

3) $345_{10} = 531_8$







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Octal to decimal, last page ques:

① 73_8

8^1 8^0

$= 56 + 3 = 59_{10}$

② 170_8

8^2 8^1 8^0

$64 + 56 = 120_{10}$

③ $531_8 =$

8^2 8^1 8^0

$320 + 24 + 1 = 345_{10}$

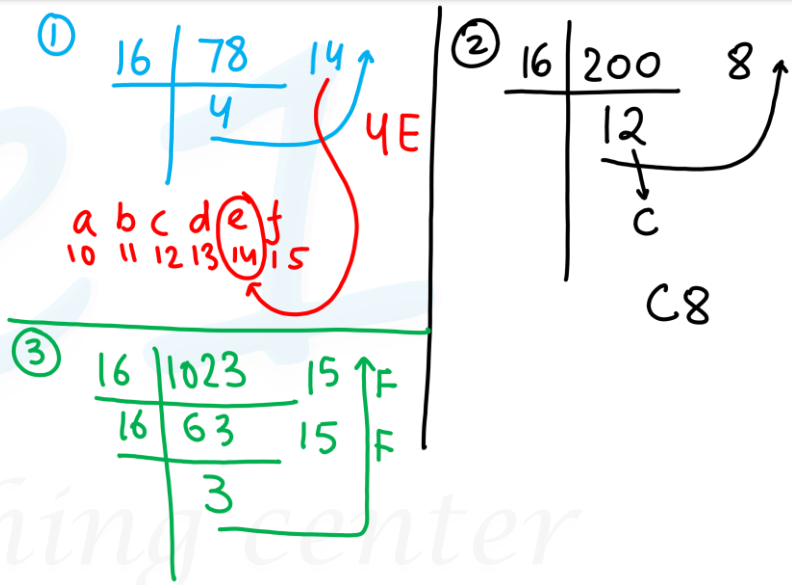
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Decimal to Hexadecimal and vice versa

1) $78_{10} = 4E_{16}$

2) $200_{10} = C8_{16}$

3) $1023_{10} = 3FF_{16}$



coaching center

a b c d e f
10 11 12 13 14 15

Hexadecimal to decimal, previous slide ques:

① $4E_{16}$ (14)

$4 \times 16^1 \quad E \times 16^0$

$= 64 + 14 = 78_{10}$

② $C8_{16}$ (12)

$C \times 16^1 \quad 8 \times 16^0$

$= 192 + 8 = 200_{10}$

③ $3FF_{16}$ (15 15)

$3 \times 16^2 \quad F \times 16^1 \quad F \times 16^0$

$16^2 = 256$

$768 + 240 + 15$

$= 1023_{10}$

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Convert the following numbers to binary, octal and hexadecimal numbers:

1) 25_{10}

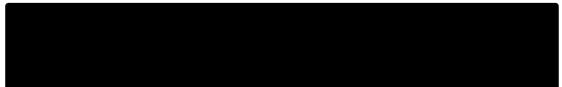
2) 58_{10}

3) 127_{10}

4) 255_{10}

5) 345_{10}

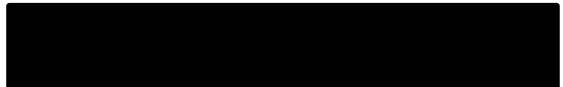
e1 coaching center



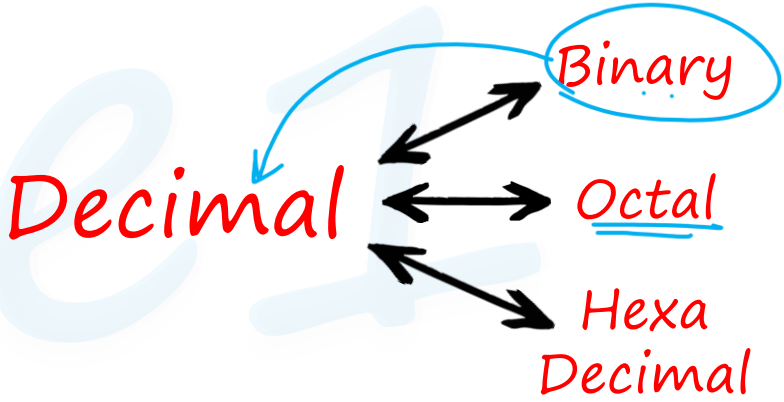
Answer key for last question:

Decimal (base-10)	Binary (base-2)	Octal (base-8)	Hex (base-16)
25	11001	31	19
58	111010	72	3A
127	1111111	177	7F
255	11111111	377	FF
345	101011001	531	159

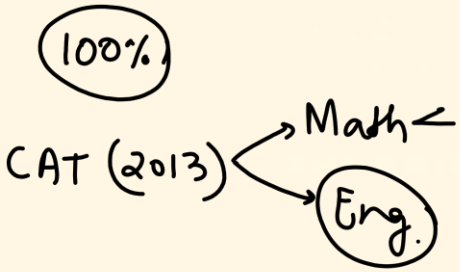
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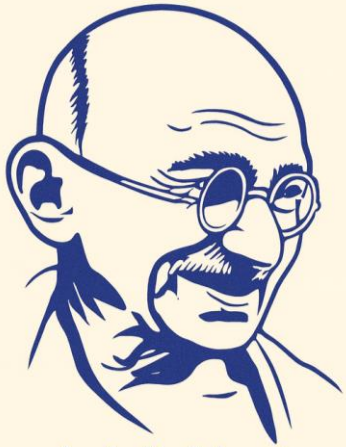


What if I have to convert from binary to octal etc:



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"If I have the belief that I can do it, I shall surely acquire the capacity to do it even if I may not have it at the beginning."